Question 28

Aim:

To verify whether a measured SQNR of 42 dB for an 8-bit ADC quantizing a sinusoidal signal is theoretically possible, and to explain the reason for any discrepancy from the ideal value.

Objective:

To compute the theoretical SQNR of an ideal 8-bit ADC using the standard quantization noise formula, compare it with the given measured SQNR of 42 dB, determine the effective number of bits (ENOB), and identify the practical factors responsible for the gap between theoretical and measured values.

Given: N = 8 bits, Measured SQNR = 42 dB

Solution:

SQNR Verification – 8-bit ADC

Given: $N = 8$ bits, Measured SQNR = 42 dB

Step 1: Theoretical SQNR formula

$SQNR(dB) = 6.02\,N + 1.76$

$= 6.02(8) + 1.76 = 48.16 + 1.76$

$\boxed{SQNR(theoretical) \approx 49.92 \text{ dB} \approx 50 \text{ dB}}$

Step 2: Discrepancy

$\Delta SQNR = SQNR(theo) - SQNR(meas)$

$= 49.92 - 42$

$\boxed{\Delta SQNR \approx 7.92 \text{ dB} \approx 8 \text{ dB}}$

Step 3: Effective No. of Bits

$ENOB = (SQNR(meas) - 1.76) / 6.02$

$= (42 - 1.76) / 6.02 = 40.24 / 6.02$

$\boxed{ENOB \approx 6.68 \text{ bits}}$

Step 4: Is it possible?

Since $42 \text{ dB} < 49.92 \text{ dB} \rightarrow$ YES, possible.

Measured SQNR can never exceed the ideal value.

Reasons for the gap

- Input signal not at full-scale amplitude
- Extra thermal / jitter noise in ADC
- ADC nonlinearity (INL / DNL errors)
- FFT spectral leakage in measurement

Conclusion

42 dB is valid. ADC performs like a ~6.68-bit ideal converter instead of 8-bit.